PCOS Symptoms

Medical Records vs. Patient Voices

MAY 03, 2026

PCOS

Also known as Polycystic Ovary Syndrome, is a common hormonal disorder that affects women of reproductive age. It is also a leading cause of female infertility and is associated with metabolic issues.

While it affects 1 in 10 women, it remains one of the most misunderstood and misdiagnosed hormonal conditions.

Research Question

Do medical PCOS symptoms match patient experiences on Reddit?

Using lifestyle factors to provide additional context

02 Data Sources

Medical Study

541 patients

To identify medically recorded observable symptoms and test which symptoms predict PCOS.

Observable symptom categories used:

1. **PCOS (Y/N)**
2. **Cycle irregularity**
3. **Cycle length**
4. **Weight gain**
5. **Hair growth**
6. **Skin darkening**
7. **Hair loss**
8. **Pimples**

r/PCOS forum

300 reddit posts only

To understand which PCOS symptoms are most discussed in real-life patient experiences.

Analyzed:

- Post titles + text
- Symptom keyword mentions
- Patient-reported observable symptoms
- Discussion frequency by symptom category

Lifestyle Study

173 patients

To explore how daily lifestyle behaviors may provide additional context for PCOS management.

Focused lifestyle factors:

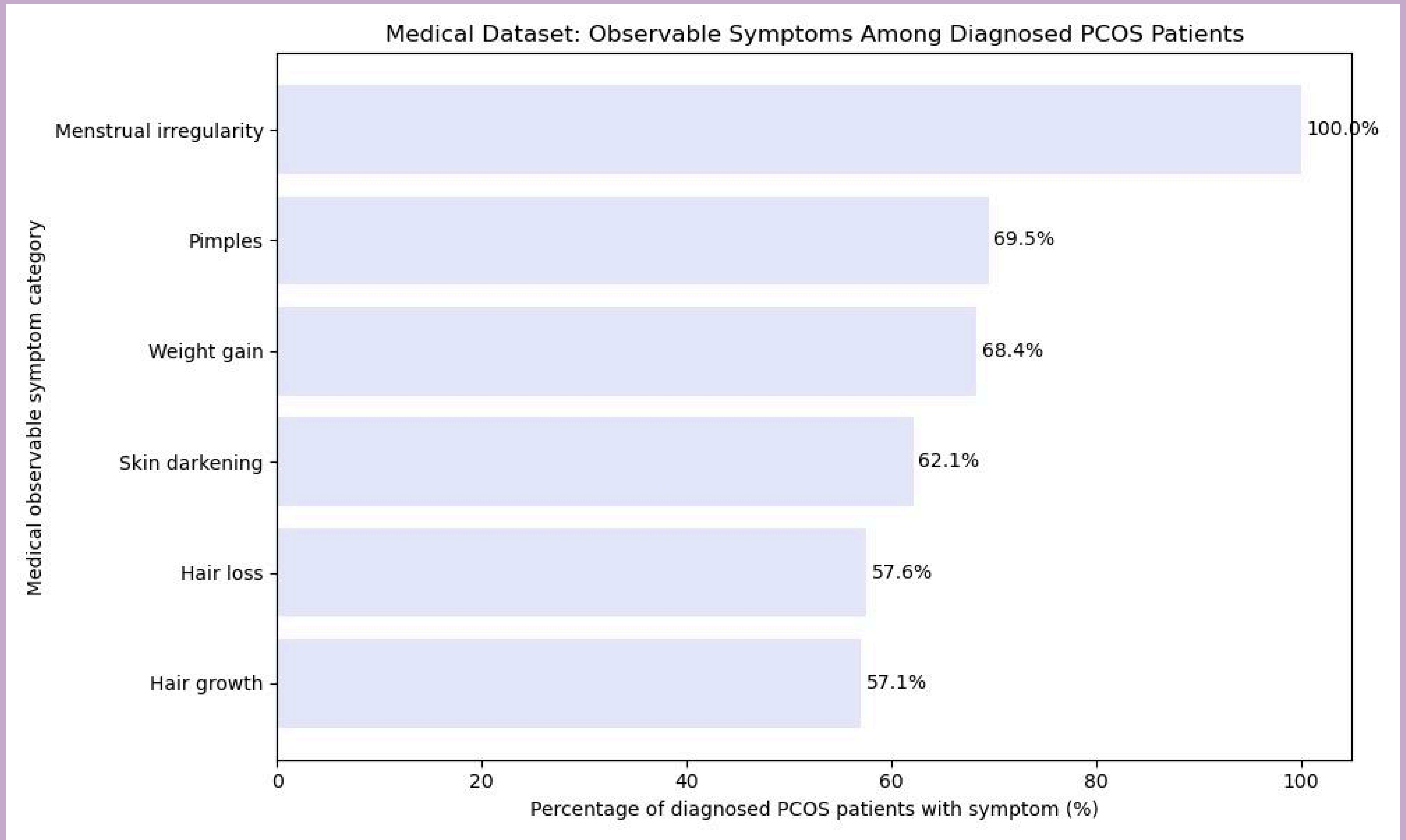
1. Stress_Level
2. Exercise_Frequency
3.Diet_Sweets
4. Diet_Fried_Food

03 Observable Symptoms

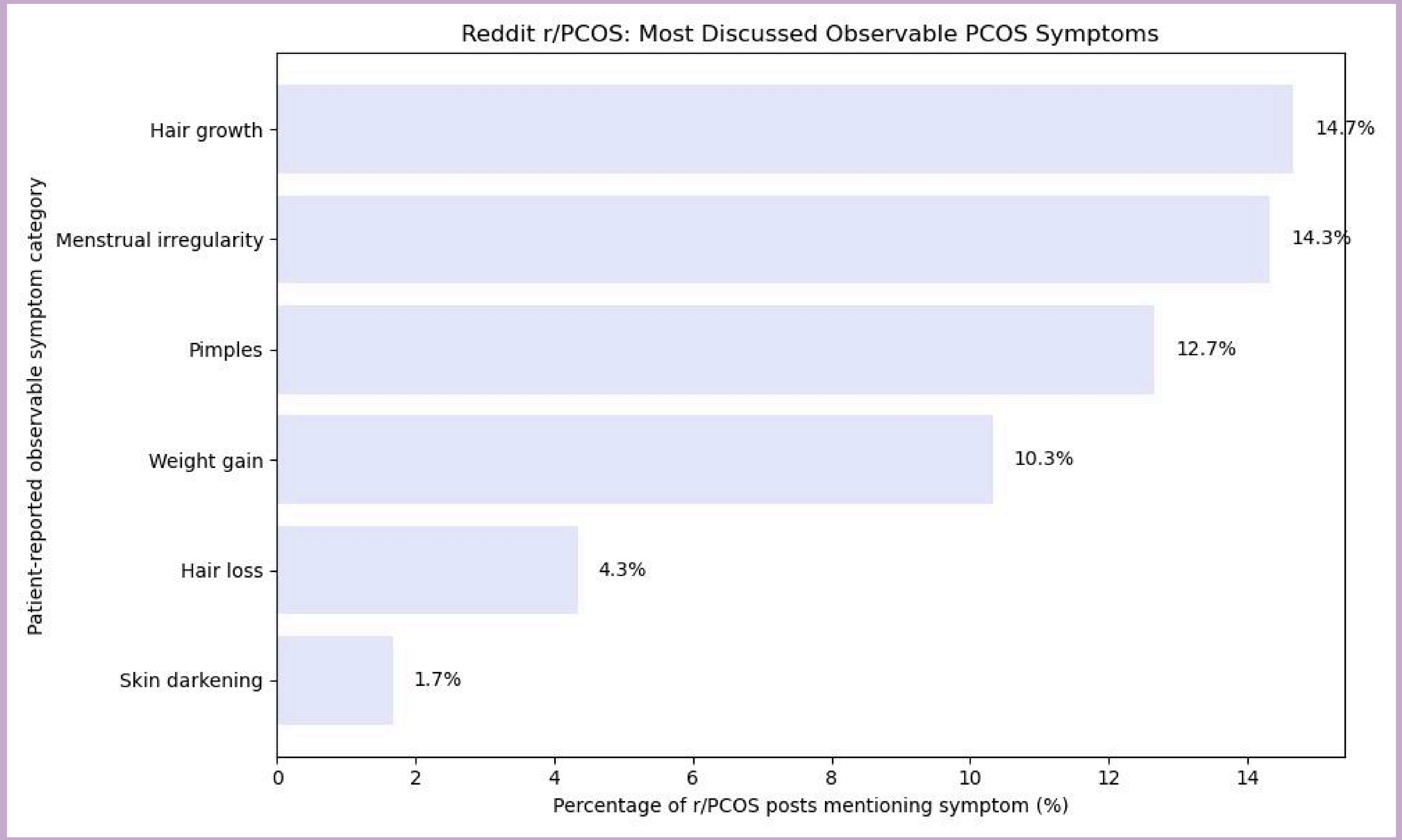
The medical dataset shows which symptoms were most common among diagnosed PCOS patients. The r/PCOS data shows which symptoms patients talked about most online.

The symptoms overlap, but the ranking is different: menstrual irregularity was highest in the medical dataset, while hair growth was highest in r/PCOS discussions.

Medical Dataset



r/PCOS forum

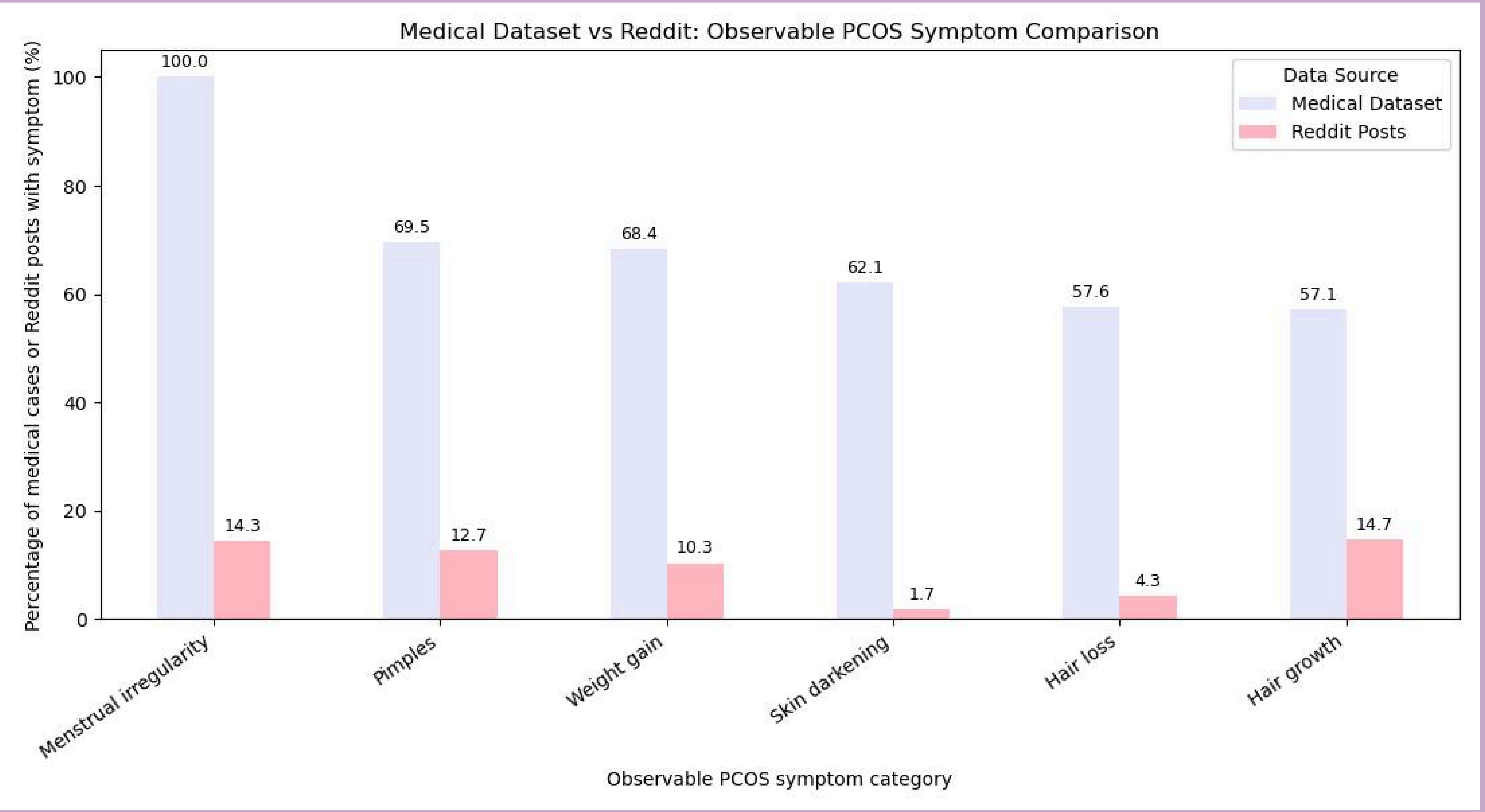


This means medical records and patient communities may emphasize different parts of the PCOS experience.

04 Medical vs Reddit Comparison

Direct comparison is limited because the medical dataset measures diagnosed patients, while Reddit measures symptom discussion in posts. However, the contrast is still useful: hair growth was one of the least observed symptoms in the medical dataset, but it was the most discussed symptom in r/PCOS.

Medical vs Reddit Dataset



This suggests that some symptoms may carry more emotional or daily-life weight for patients, even if they are not the most common symptoms recorded medically.

05

From Symptom Comparison to Prediction

- [x] Compared medical symptom prevalence
- [x] Compared Reddit symptom discussions
- [x] Found different symptom emphasis
- [ ] **Tested whether symptoms can predict PCOS**
- [ ] **Compared classifier performance**

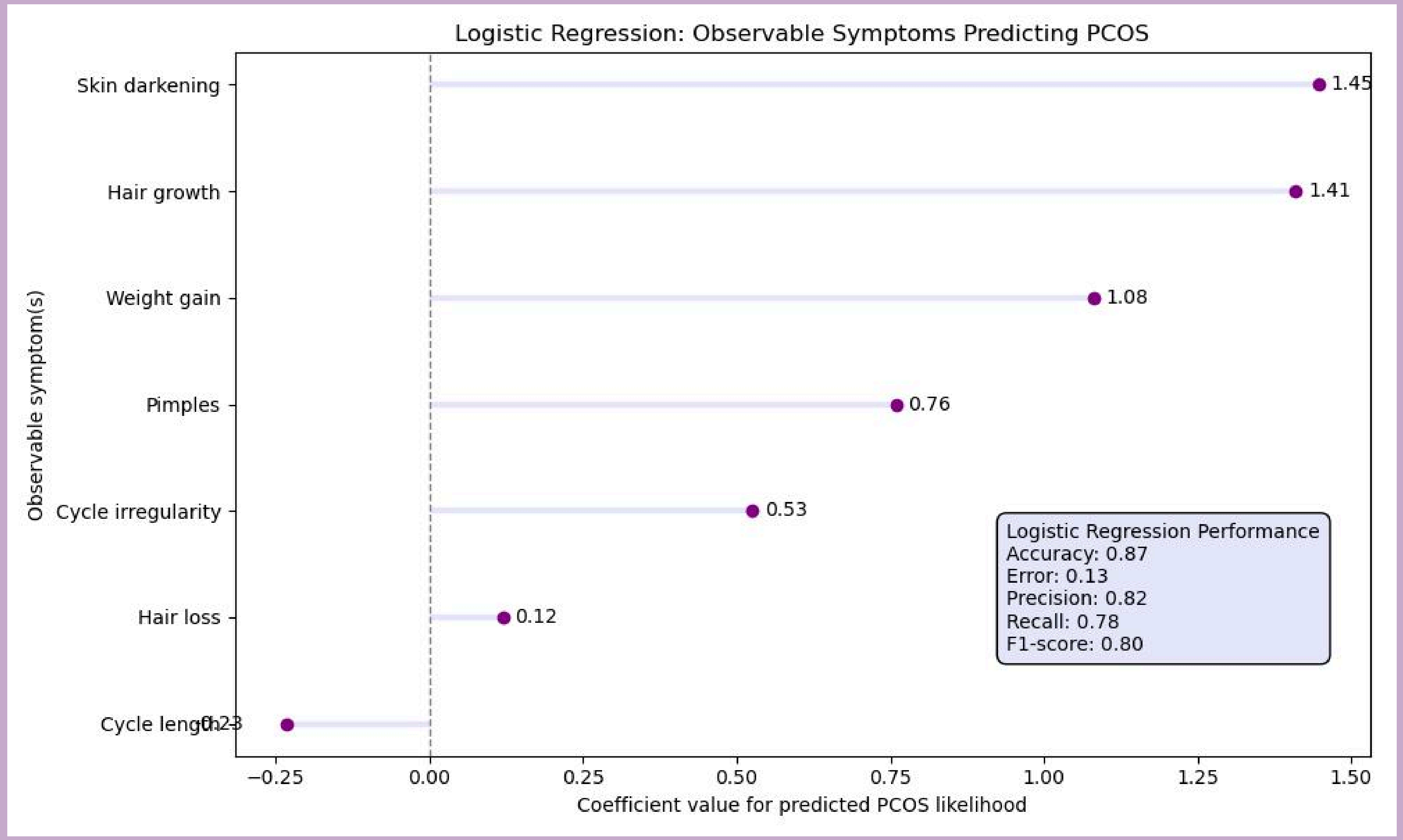
I tested whether the same observable symptoms could help predict PCOS diagnosis in the medical dataset. **Performance scores show how well each model predicted PCOS, while coefficient and importance scores show which symptoms influenced the prediction.**

*Note: I'm currently taking DSCI552: Machine Learning as well.

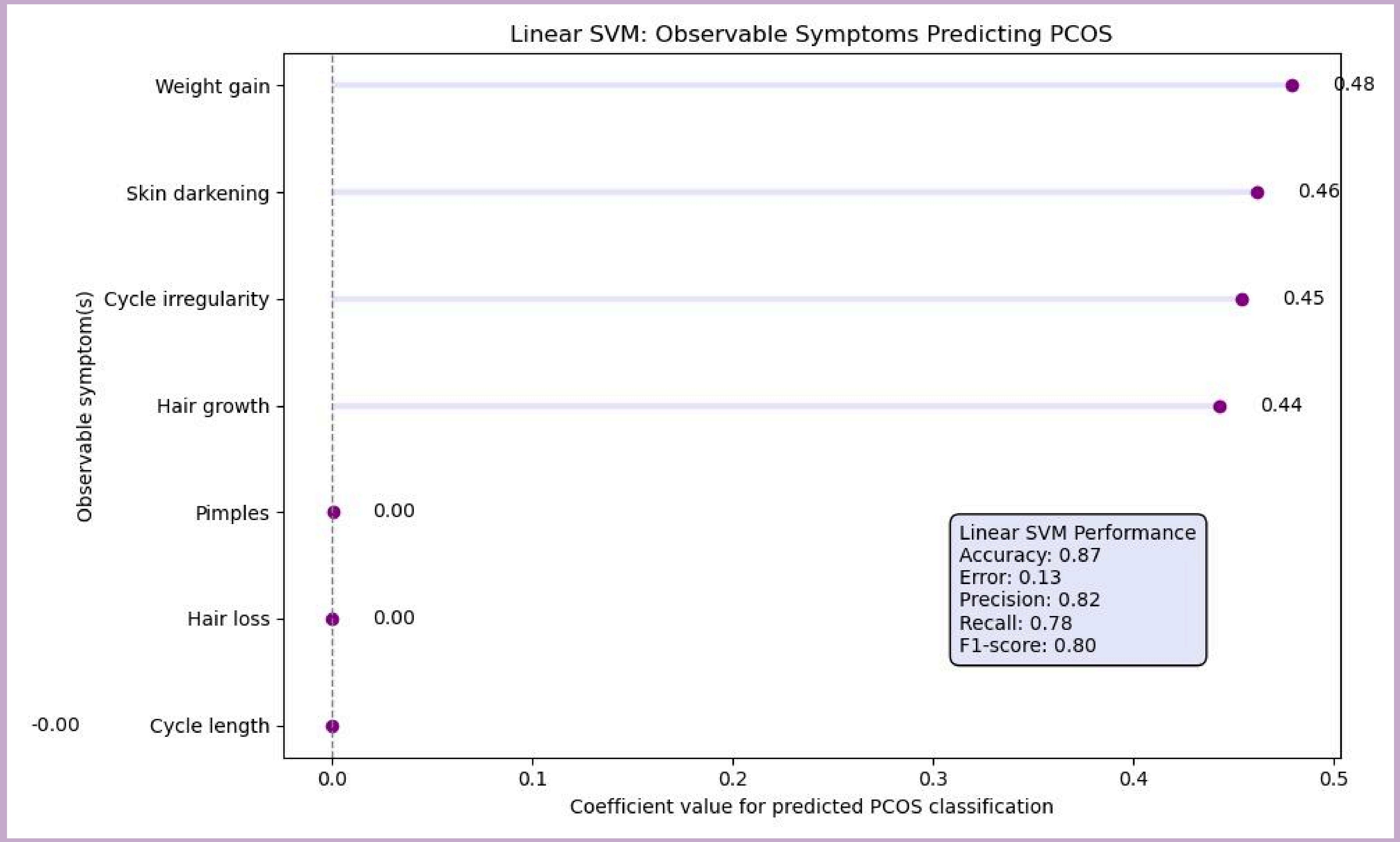
Baseline Classifiers Show Similar Predictive Performance

I first tested Logistic Regression and Linear SVM because both are simpler, interpretable models that help establish a baseline before comparing more complex tree-based classifiers..

Logistic Regression



Linear SVM



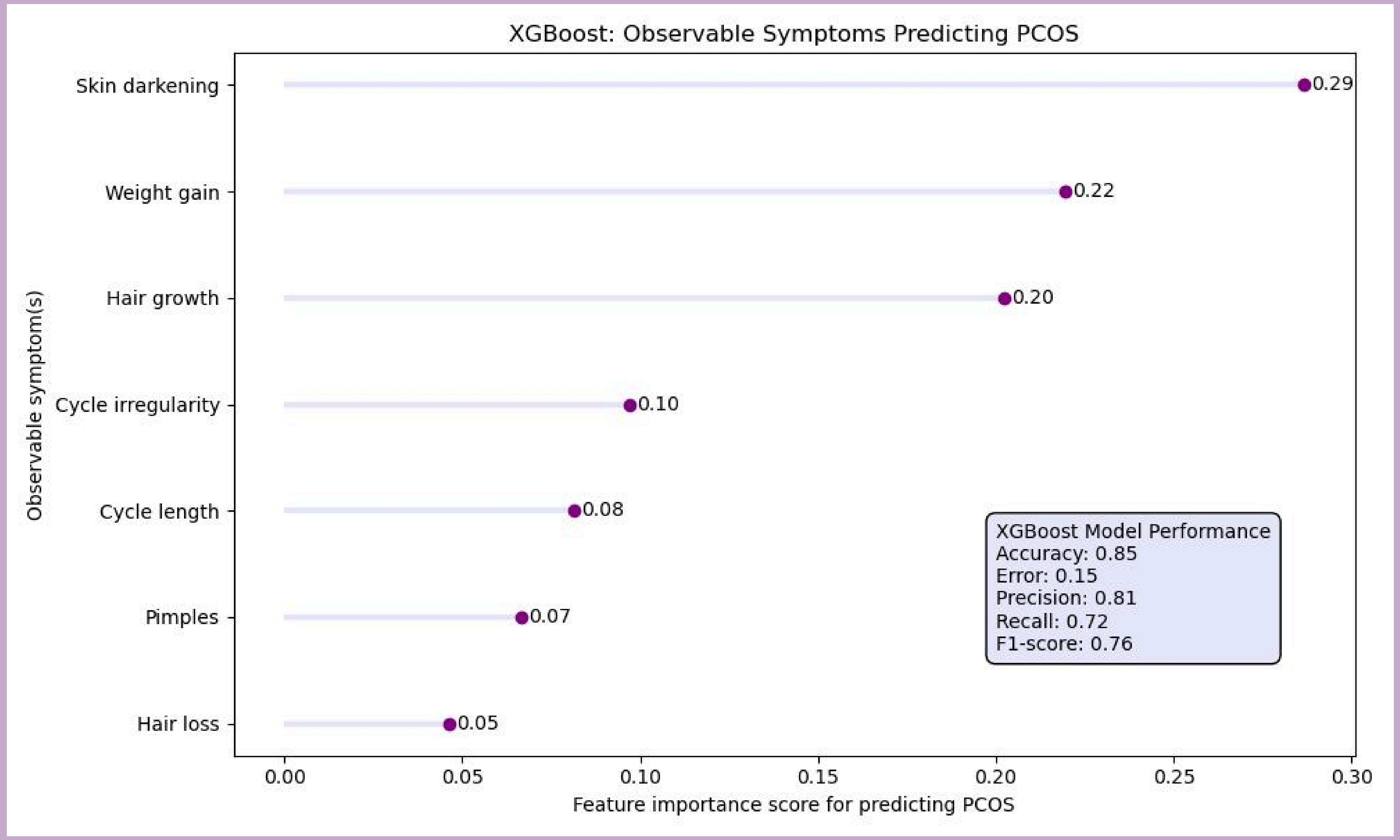
Logistic Regression: skin darkening, hair growth, and weight gain had the strongest positive coefficients.
Linear SVM: weight gain, skin darkening, cycle irregularity, and hair growth carried most of the predictive weight.

Even though Logistic Regression and Linear SVM had the same accuracy, precision, recall, and F1-score, their coefficient charts show that they weighted symptoms differently.

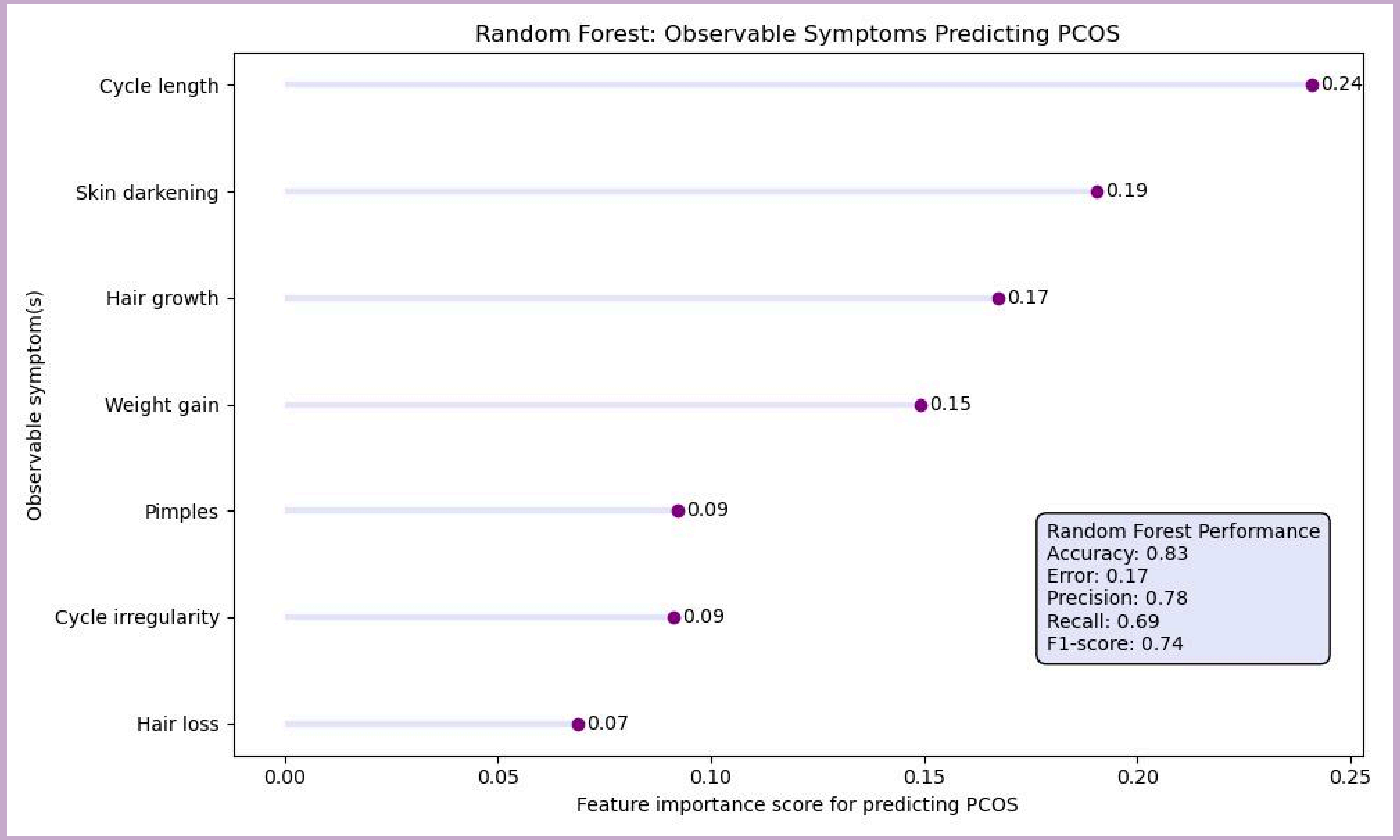
Tree-Based Models Highlight Different Predictive Symptoms

Then, I used XGBoost and Random Forest to see whether tree-based classifiers could capture more complex symptom patterns. XGBoost had higher accuracy and F1-score, while Random Forest placed the most importance on cycle length.

XGBoost



Random Forest



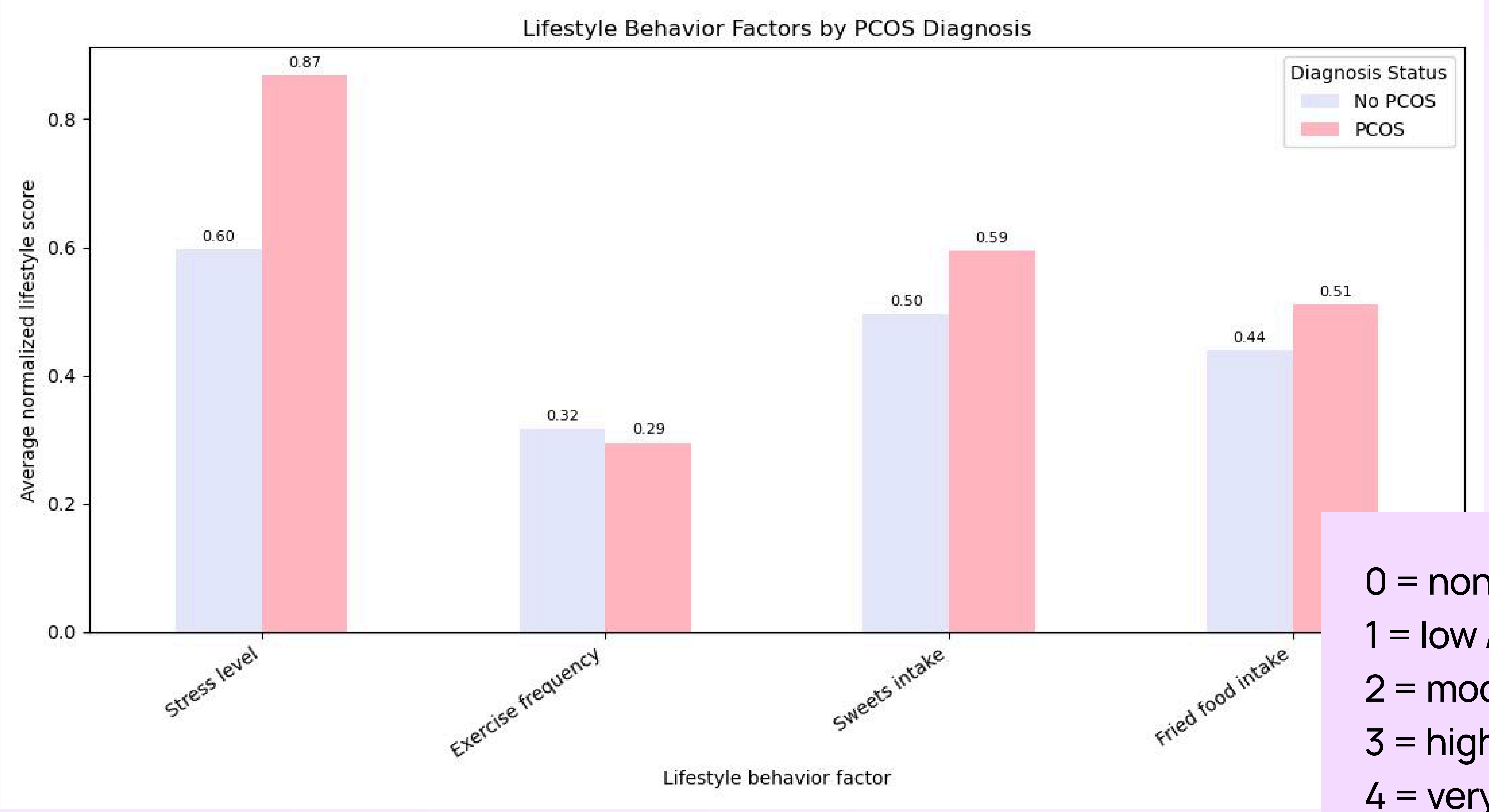
Both tree-based models show that observable symptoms can help predict PCOS, but they ranked symptom importance differently. They identified skin darkening and hair growth as important predictors, suggesting these symptoms carry meaningful predictive value in the medical dataset.

Beyond observable symptoms, what lifestyle factors may be associated with PCOS?

I focused only on lifestyle behavior factors: stress, exercise, sweets intake, and fried food intake. These are not symptoms, so they were analyzed separately from the Medical vs Reddit symptom comparison. Because the lifestyle columns may use different scales, I normalized them from 0 to 1 so they could be compared in one chart.

06 Are lifestyle factors like stress, diet, and exercise different between people with and without PCOS?

People with PCOS showed higher average normalized scores for stress, sweets intake, and fried food intake, while exercise frequency was slightly lower.



0 = none / never / no
1 = low / rarely / yes
2 = moderate / sometimes
3 = high / often / regularly
4 = very high / daily

In conclusion...

Medical symptoms and patient experiences partially overlap, but they are not emphasized the same way. Adding lifestyle context gives a fuller picture of PCOS beyond diagnosis alone.

Main Finding

PCOS symptoms appear differently across medical records, patient discussions, and lifestyle data.

What I learned

Medical data shows what is recorded. Reddit shows what patients talk about. Lifestyle data shows daily factors that may shape management.

PCOS care needs data, context, and empathy.

08

Challenges

🔴 Google Trends	Many errors, had to change my source (too late)
🔴 r/PCOS:Scope	Realized I had to narrow my project to analyzing symptoms that were negative ONLY, and filtering it (many data sets)
🔴 .main & .config	Putting everything together, DEBUGGING